IPCV · SEMESTER VII · TEAM 25

QuickPic

One photo. The whole register.

Open-set face recognition for classroom attendance from a single group photograph



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Roll call takes minutes. One class photo should be enough, but three things make it hard.

30 px

Tiny faces

Back-row faces are a few dozen pixels tall, far below what detectors and recognisers are tuned for.

~85 %

Mostly strangers

Most faces in our photos belong to people who are not enrolled. The system must say “unknown”, not guess.

69–102

Real conditions

Faces per photo, with turned heads, uneven light, and photos taken two years before enrollment.

Frozen pretrained networks, plus our own classroom-aware decision layer

Detect
SCRFD, 5 landmarks

Tiling
ours

Align
112 × 112

Embed
ArcFace, 512-d

Match
degraded gallery · ours

Assign
Hungarian · ours

Why not retrain?

ArcFace learned from hundreds of thousands of people. Retraining on a few hundred Indian faces would likely make it worse, and needs a GPU.

What we built instead

Three classroom fixes around the networks, each switchable, so we can measure exactly what each one adds.

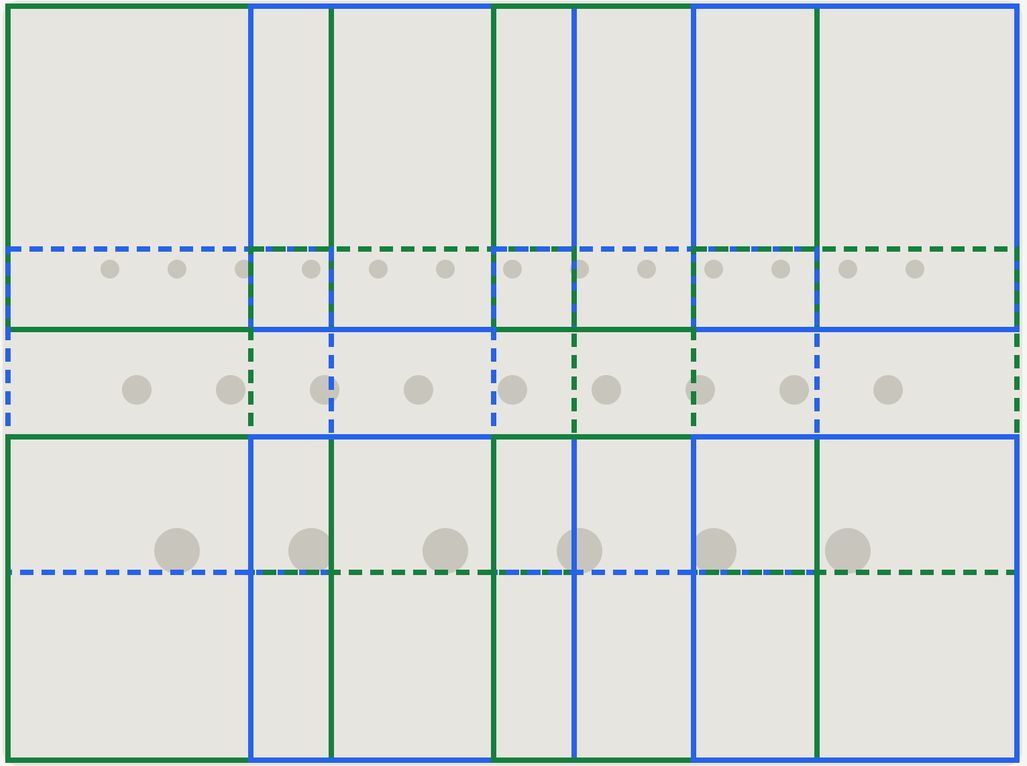
Tiling: look at the photo at full resolution

Detectors shrink every photo to 640 px. In a 4032-px class photo, a 40-px face becomes **6 px** and vanishes.

1. Cut the photo into overlapping 1280-px tiles (25 % overlap) and detect in each, plus one pass on the whole photo.

2. Drop boxes cut by a tile edge; merge duplicates with non-maximum suppression.

3. Smallest face found reliably: **96 px → 28 px**.



A 4032 × 3024 phone photo covered by 12 overlapping 1280-px tiles

Tiling raises recall for every detector we tried

Detector	Whole photo	With tiles	ms / photo
SCRFD-10GF (used)	87.9 %	**97.4 %**	1167
SCRFD-500MF	80.5 %	94.6 %	135
YuNet	83.4 %	96.3 %	102
Haar cascade	5.0 %	74.8 %	347

5 → 75 %

Even the classical Haar detector (2001) goes from almost nothing to three faces in four.

YuNet is close to SCRFD but adds 66 false boxes, so we keep SCRFD.

Recall on 793 hand-labelled faces in 10 real class photos (4032 × 3024). CPU time on an Apple M2 laptop.

SHASHANK · GROUND TRUTH

Building the answer key: every face in 10 photos, labelled by hand

01 **Propose every possible face**

Our labelling tool runs tiled SCRFD at a low threshold (0.3), so almost nothing is missed.

02 **Tag each box**

An enrolled student’s roll number, “unknown”, or “not a face”. Click the photo to add a face the detector missed.

03 **Score against it**

A detection counts as a face when it overlaps a labelled box by IoU ≥ 0.4. Runs locally; no photo leaves the laptop.

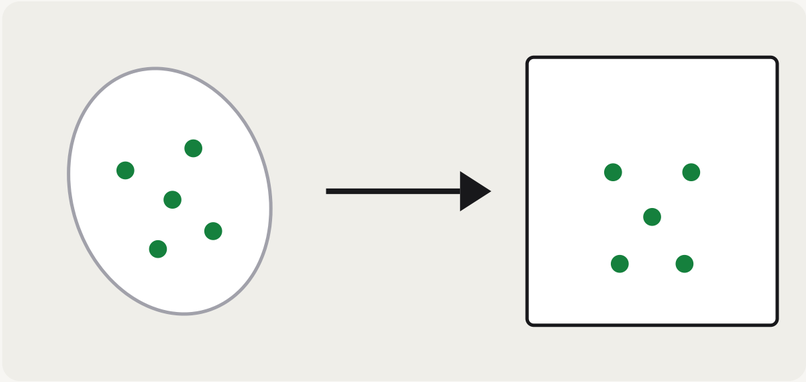
804
faces labelled

117
student appearances

676
strangers to reject

11
false boxes, not faces

Image processing prepares every face



Five landmarks are mapped onto a fixed template by a **least-squares similarity transform** (rotate, scale, shift): an upright 112 × 112 crop.

Quality checks on registration photos

Check	How it is measured
Sharpness	Variance of the Laplacian ≥ 15
Lighting	Mean brightness 45–215
Size	Face at least 120 px tall
Head turn	Nose offset ÷ eye distance

The head-turn estimate tells “facing the camera” (under 0.15) from “turned to one side” (0.12–0.8), with no extra model.

Two fixes so that the right student gets the name

Degraded enrollment

Registration selfies are sharp; classroom faces are small and blurry. So each student is stored with 10 extra copies of every photo, made to look like a back-row face.

original | shrunk to 24–48 px | blurred | darker / brighter | JPEG | flipped

One face per student

Faces are named jointly with the Hungarian method: highest total similarity, each student used once, scores under 0.35 stay unknown.

score	S1	S2	assigned
Face 1	**0.62**	0.31	S1
Face 2	0.41	**0.38**	S2, not S1 again
Face 3	0.29	0.12	unknown

When two students look alike, it asks the teacher instead of guessing

If the second-best student scored almost as high as the best (gap under **0.06**), the name is shown in **amber with “?”** on the review screen.

example	score	gap	shown
correct	0.38	0.071	green
correct	0.40	0.073	green
wrong	0.37	0.054	amber ?
wrong	0.36	0.053	amber ?

2 of 2

borderline mistakes flagged for review

2 of 115

correct names flagged unnecessarily (a plain score cut-off flagged 6)

Four datasets, one honest test protocol

Dataset	What it contains	What it tests
LFW	13,233 web photos of 5,749 people	Clear faces; calibration training
TinyFace	169k natively tiny faces, about 20 × 16 px	Rejecting strangers on tiny faces
ICFD (IIT Madras)	Indian students filmed in real classrooms	Indian faces without retraining
Our class	10 photos, 804 faces we labelled, 15 enrolled	The real task, end to end

Leave one photo out

The cut-off is tuned on 9 photos and tested on the 10th, never on the photo being scored.

Metrics that matter

Attendance F1, students named correctly, and strangers wrongly named.

Error bars

95 % intervals from 2,000 bootstrap resamples of the photos.

Near-perfect on clear faces, much harder on tiny ones

Face model	LFW	TinyFace	TinyFace open-set	Indian classroom
ArcFace ResNet-50 (used)	**99.8 %**	**43.3 %**	15.0 %	**98.5 %**
ArcFace MobileFaceNet	99.5 %	36.9 %	**21.0 %**	98.0 %
SFace	99.4 %	26.1 %	17.2 %	95.5 %
LBP histograms (classical)	58.5 %	20.0 %	16.4 %	8.0 %

No retraining needed for Indian faces

98.5 % on real classroom footage from IIT Madras, with the model as it is.

Calibration: an honest negative

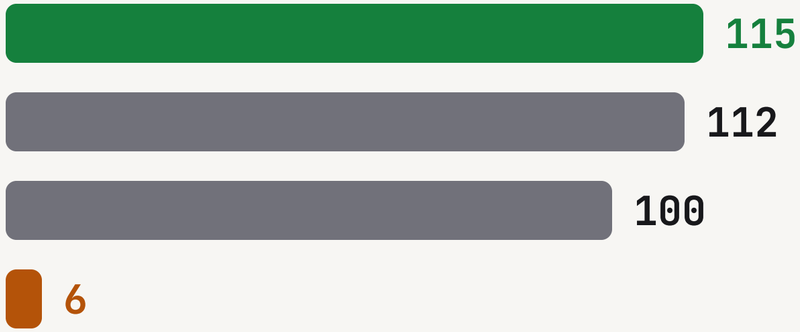
Our score calibration lifts 12-px faces from 86 % to 93 % accepted, but does not help at classroom sizes, so it is off.

LFW: verification · TinyFace: rank-1 and open-set (1 % false alarms) · Indian: ICFD rank-1

UTKARSH · FACE MODELS ON OUR CLASS

On our own class, ArcFace ResNet-50 is the clear choice

FACE MODEL	NAMED CORRECTLY (OF 117)	STRANGERS
ResNet-50 (used)	115	6
MobileFaceNet	112	6
SFace	100	15
LBP (classical)	6	121



121

strangers named by the classical LBP method

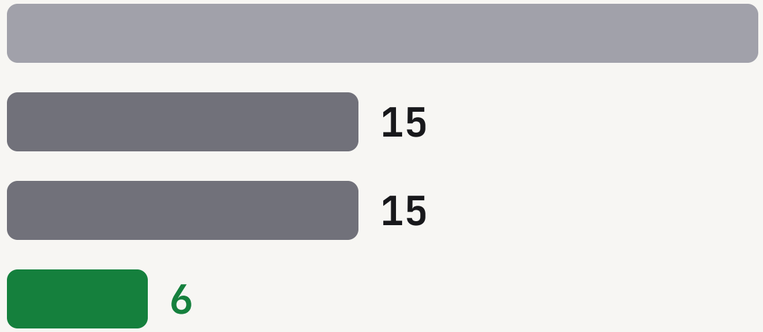
Hand-crafted texture features do not survive real classroom photos; learned features do.

All four models run inside the same final pipeline (tiling, degraded enrollment, one-to-one), on the same 10 photos.

TANISH · RESULTS ON OUR CLASS

Each fix removes mistakes: 5× fewer strangers given a student's name

CONFIGURATION	STRANGERS NAMED	F1
Pretrained baseline	32	0.954
+ tiling	15	0.954
+ degraded enrollment	15	0.966
+ one-to-one (final)	**6**	**0.966**



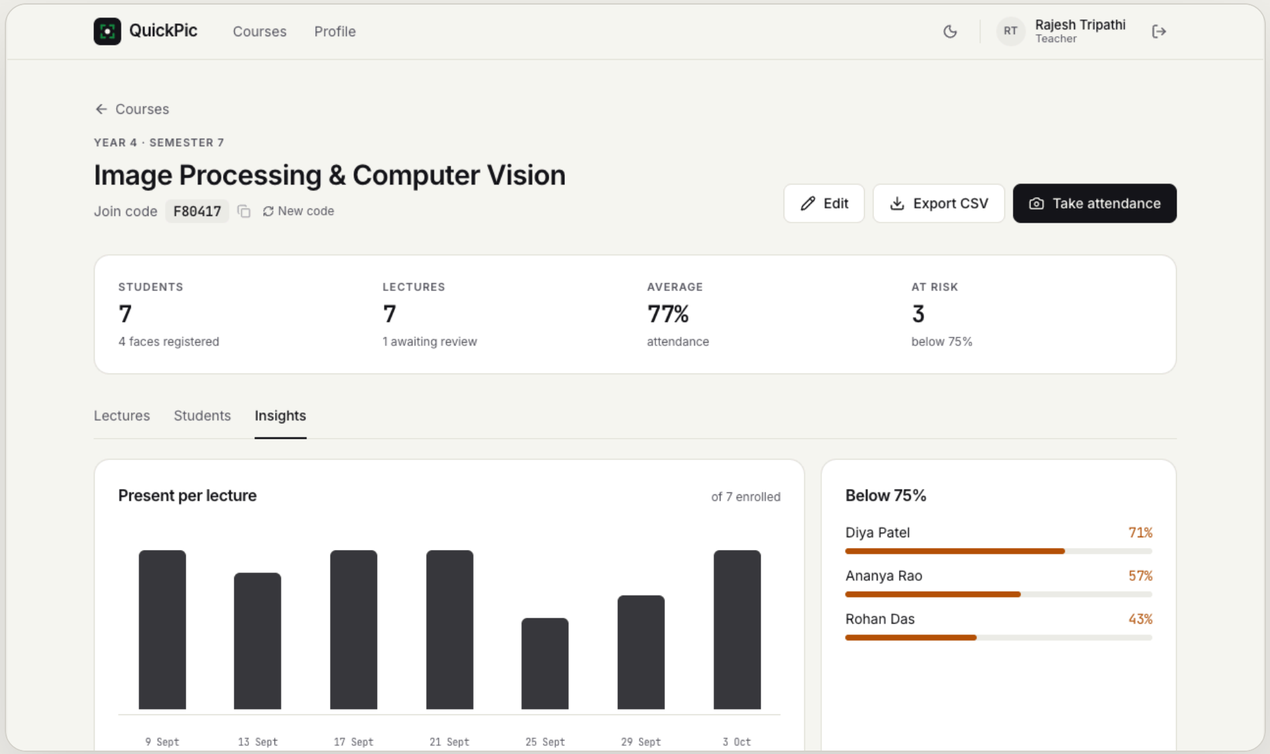
115/117

student appearances named correctly

0.8 teacher corrections per photo, down from 1.1

10 real class photos · 804 hand-labelled faces · 15 enrolled students · threshold never tuned on the photo being scored

The system drafts the register; the teacher confirms it



1 · Create a course

Students join with a six-letter code.

2 · Upload one class photo

Every face is found and named in about a second on the laptop GPU.

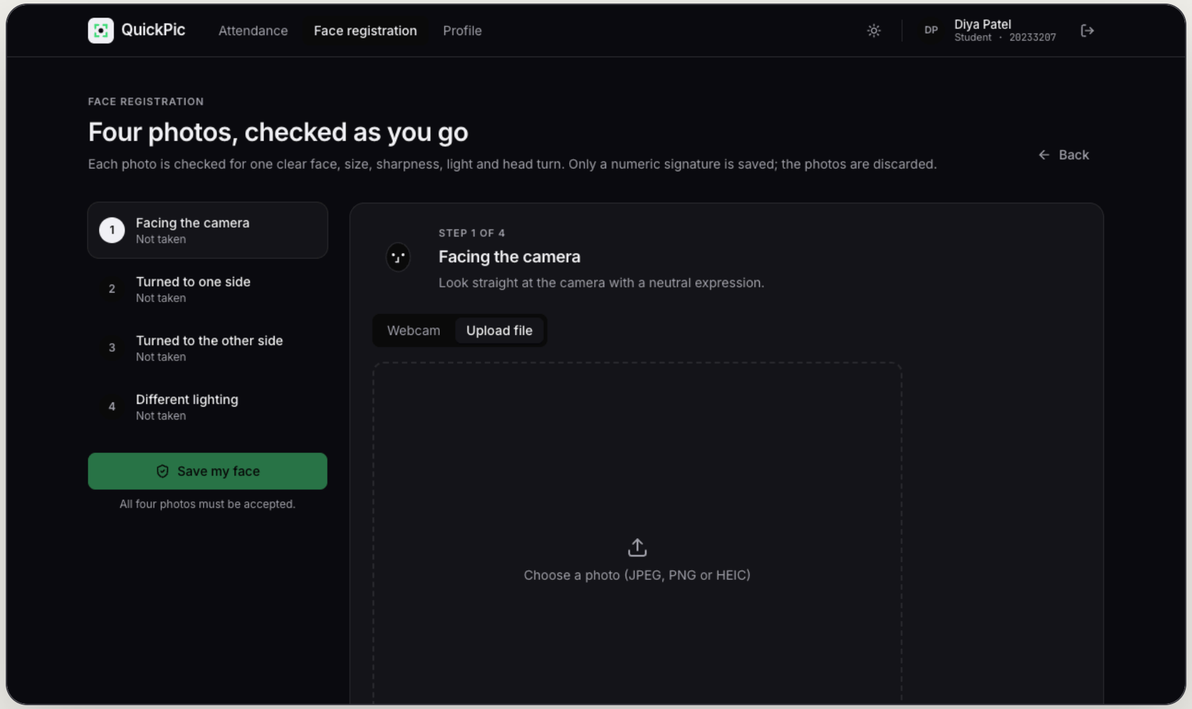
3 · Review on the photo

Green named, **amber ?** unsure, dashed unknown. Fix any face with a click.

4 · Confirm

The photo is deleted. Insights, edits and CSV export stay.

Students register themselves; privacy is built in



Four guided photos

Front, left, right, other light; webcam or file. Each is checked instantly: one face, big enough, sharp, evenly lit, head turned as asked.

Signatures, not photos

Only face signatures are stored. Class photos are deleted on confirm. Students can delete their data any time. Nothing leaves the laptop.

From research code to a working product

React app · TypeScript
Review screen: zoomable photo, editable face boxes, one-face-per-student rule

Django JSON API · typed, CSRF-protected
Accounts, courses, lectures, registration checks, attendance

Recognition engine · one Python package
The same code runs every experiment in the report and the live website

6 s → 0.5 s
per class photo on the laptop GPU (Apple CoreML), same names as on CPU

58 tests
run automatically on every push, plus a guard that blocks any face photo from the public repo

LIVE DEMO

Let’s take attendance.

01

A teammate registers live with the webcam

02

The teacher uploads a class photo

03

Review, fix one face, confirm

What we would do with more time

Limitations

Small test set: 15 enrolled students, 10 photos.

Faces under about 25 px are still missed or unrecognised.

Lookalikes need a teacher’s check.

Face boxes were labelled starting from one of the tested detectors.

Next steps

Enroll the whole class and test on more lectures.

Learn from confirmed attendance: add each confirmed classroom face to the student’s gallery.

Fit calibration on classroom faces instead of LFW.

Use seating patterns across lectures as extra evidence.

TEAM 25 · WHO DID WHAT

Thank you. Questions?

Tanish Mittal

20233288 · lead

System design, recognition engine, website architecture and review screen, GPU speed-up, evaluation, report

Shashank Kanaujiya

20233259

Face detection, tiling, detector comparison, labelling tool and ground truth, alignment, image-quality checks

Utkarsh Raj Singh

20233298

Datasets and evaluation protocol, public benchmarks, face-model comparison, calibration study

Shinde Ritesh Tirupati

20233350

Website registration and dashboard pages, testing, photo collection, literature survey

github.com/Tanish-237/quickpic